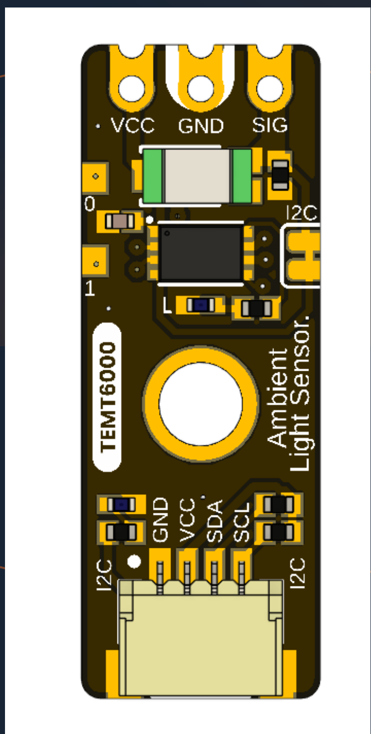
PRODUCT REFERENCE MANUAL

UNIT ELECTRONICS · DEVLAB ECOSYSTEM · ATOM FAMILY

# UNIT ATOM TEMT6000

Version 1.1.0




TEMT6000 ambient-light module with an onboard I2C interface, Qwiic expansion, and direct analog access.

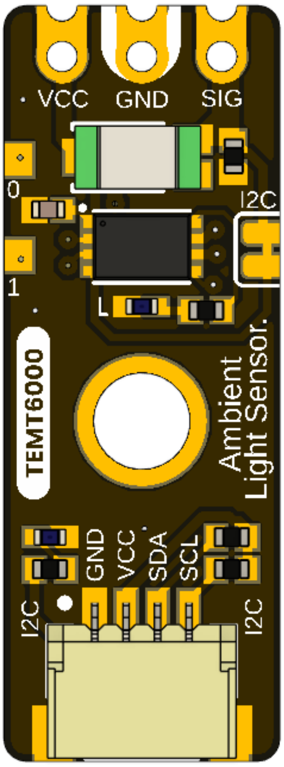
TEMT6000 | I2C + ANALOG | QWIIC | DEVLAB ECOSYSTEM | ATOM FAMILY

DOCUMENT MAP

# Contents

# Description

The UNIT ATOM TEMT6000 is an I2C-compatible ambient-light sensor module from UNIT Electronics, the company that creates and develops the DevLab board ecosystem. Atom is a product family within that ecosystem. Hardware V0.3.1 combines a TEMT6000 visible-light phototransistor with an onboard interface controller, Qwiic I2C routing, and direct access to the sensor's analog signal. The fitted sensor manufacturer and exact orderable suffix have not been confirmed.




The released pinout identifies two optional Qwiic connection positions, direct `VCC`/`GND`/`SIG` contacts, reserved `PA0` and `PA1` pads, factory-only reset and SWD functions multiplexed on the I2C port, power and built-in indicators, and a solder bridge that disables the I2C function when cut. Inside the controller, the sensor ADC input is PA2, `BUILTIN` is driven by PB5, and the I2C bus uses `PB6/SCL/SWCLK` and `PA10/SDA/SWDIO`.

## Applications

- I2C-connected ambient-light acquisition
- Direct analog light measurement during development and calibration
- Automatic display and indicator brightness control
- Day/night and relative-light detection
- Environmental data logging
- Educational I2C, ADC, and phototransistor experiments

## DevLab Format Compatibility

As part of the DevLab Atom family, the TEMT6000 module uses a compact, standardized format for rapid prototyping and integration into sensing systems. It provides direct access to the analog signal through the VCC, GND, and SIG contacts, while the onboard controller publishes digital sensor readings through I²C. The module is intended for nominal 3.3 V or 5 V operation. The controller guidance uses a conservative 2.0–5.5 V range; complete-module electrical limits require the current schematic and validation.

The board has two optional positions for horizontal 4-pin, 1.00 mm-pitch JST/Qwiic connectors carrying GND, VCC, SDA, and SCL. They provide alternative I²C connection points. Before connection, confirm the populated connector and its orientation, and ensure the host tolerates the actual I²C pull-up voltage; a 3.3 V host may require level translation when the module is powered at 5 V. The pull-up resistance is not specified by the released documentation.

## Hardware Features

- TEMT6000 ambient-light phototransistor; fitted manufacturer/orderable part not confirmed
- 32-bit Arm Cortex-M0+ controller with 16 KB Flash and 2 KB SRAM, using its internal HSI at up to 24 MHz; no external oscillator is used
- Nominal 3.3 V and 5 V operation; controller upper operating limit 5.5 V
- 7-bit I2C slave operation from 100 kHz through 400 kHz
- Qwiic GND, VCC, SDA, and SCL routing
- Direct analog SIG access with adjacent VCC and GND
- Reserved PA0 and PA1 pads with no current application assignment
- Internal sensor ADC connection to controller pin PA2
- Shared PB6 / SCL / SWCLK / PA10 / SDA / SWDIO controller mapping
- Factory-only SWD/reset functions, power LED, and PB5-controlled built-in LED
- Cuttable solder bridge for disabling I2C operation
- Manufacturer Part Number (MPN) UE0098

The controller implements DevLab Device Protocol (DDP) v1.0. The TEMT6000 profile is identified by Device ID 0x0102; command TEMT6000_RAW (0x80) returns a 12-bit ADC sample in an unsigned 16-bit little-endian response. The current controller firmware reports factory address 0x20, firmware/hardware 1.0, and capabilities 0x000001B9. Procurement does not establish the fitted controller suffix, so the published controller voltage guidance uses the conservative x7 range without identifying the physical variant. The technical package does not provide the exact controller package or ordering suffix, current-revision schematic, or complete module electrical limits.

SWDIO shares physical pin PA10 with SDA, and SWCLK shares PB6 with SCL; there is no separate SWD port and the two modes are mutually exclusive. SWD/reset are reserved for manufacturer programming and advanced factory diagnostics, not user firmware replacement.

# 1 The Board

The design supports two host paths. A Qwiic-capable controller can attach through the I2C positions, while an ADC-capable host or test instrument can sample the exposed SIG contact directly. Factory debug reuses the same controller pins and physical port as I2C; it is not a separate user interface.

## 1.1 Accessories

No accessory bundle is specified. Typical integration items are:

| Accessory | Purpose | Selection notes |
|---|---|---|
| Qwiic cable | Connects GND, VCC, SDA, and SCL | Verify 1.0 mm pitch, orientation, and contact order; connectors are shown as optional |
| I2C-capable host | Scans and communicates with the module | Use 7-bit addressing and a clock from 100 kHz through 400 kHz |
| Analog test lead or carrier | Accesses VCC, GND, and SIG | SIG must connect to a voltage-compatible ADC input |
| Logic analyzer | Checks I2C activity | Use input thresholds compatible with the powered board |
| Reference lux meter | Supports optical calibration | Required for quantitative illuminance validation |

## 1.2 Board Identification

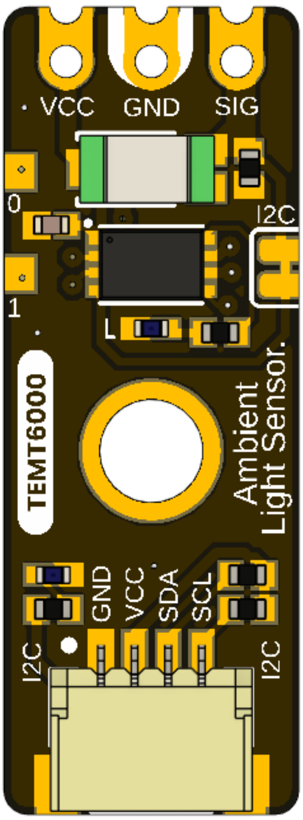
| Item | Value |
|---|---|
| Product | UNIT ATOM TEMT6000 Ambient Light Sensor |
| Brand / company | UNIT Electronics |
| Board ecosystem | DevLab |
| Product family | Atom |
| Product type | I2C-compatible and direct-analog ambient-light module |
| Optical component | TEMT6000; fitted manufacturer/orderable part not confirmed |
| Interface controller | 32-bit Arm Cortex-M0+; 16 KB Flash and 2 KB SRAM; fitted suffix not asserted |
| Manufacturer Part Number (MPN) | UE0098 |
| Current board artwork | |
| Available schematic | Legacy analog hardware V0.0.1 only |
| Product Reference | Version 1.1.0 |

Board, pinout, schematic, and documentation revisions are controlled independently.

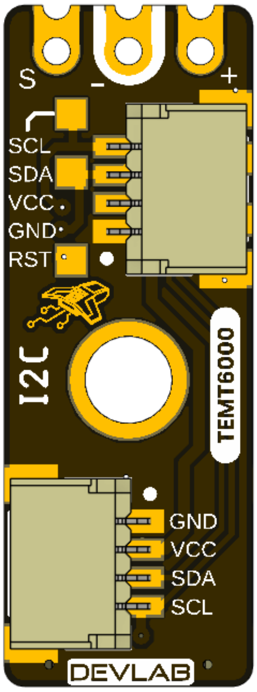
## 1.3 Identified Assemblies

| Assembly | Function | Source status |
|---|---|---|
| TEMT6000 sensor | Converts visible light to photocurrent | Functional identity shown by board/pinout artwork; manufacturer and exact suffix unconfirmed |
| Interface controller | Samples or processes the sensor for I2C access | 32-bit Arm Cortex-M0+; 16 KB Flash and 2 KB SRAM; fitted suffix not asserted; published voltage guidance uses the x7 range |
| Qwiic positions A and B | I2C power and bus access | Shown as optional horizontal JST connectors |
| Direct contacts | VCC, GND, and analog `SIG` | Identified on the top view |
| I2C disable bridge | Disconnects or disables I2C when cut | Function identified; exact circuit unspecified |
| Power and built-in LEDs | Power and firmware indication | `BUILTIN` is driven by controller PB5 |
| PA0/PA1 and service functions | Reserved GPIO and factory reset/debug | `PA0`/PA1 have no current application; SWD aliases share the I2C port |
| Internal controller mapping | PA2 ADC, PB5 built-in actuator, `PB6`/`SCL`/`SWCLK`, `PA10`/`SDA`/`SWDIO` | Current firmware/hardware mapping |

## 1.4 Board Views




The top view shows the direct sensor contacts, controller, sensor, mounting hole, indicator circuitry, and one Qwiic connector position.




The bottom view labels Qwiic, the SWD aliases on that same port, reset, and the second optional connector position.

## 1.5 Handling

Use normal ESD precautions. Keep the transparent sensor package clean and optically unobstructed. Remove power before changing connectors or modifying the I2C solder bridge. Do not attach an SWD probe or attempt to replace the firmware. SWD/reset access is reserved for the manufacturer and advanced factory diagnostics; I2C must be inactive and isolated before factory SWD use.

# 2 Ratings

No current-revision schematic or released board-level electrical table is present for V0.3.1. This chapter therefore separates the expected Qwiic integration domain from comparative TEMT6000X01 reference data and explicitly marks current module values that still require release. That external data does not identify the manufacturer or exact part fitted to the board.

## 2.1 Current Module Operating Conditions

| Parameter | Value | Status |
|---|---:|---|
| Nominal module supply | 3.3 V or 5 V | Supported operating points; use a current-limited source during bring-up |
| Published controller operating voltage | 2.0–5.5 V | Conservative x7 voltage range; supports nominal 3.3 V and 5 V without asserting the fitted suffix |
| I2C logic domain | Follows the actual bus pull-up rail | At 5 V, use a 5 V-tolerant host or bidirectional level translation |
| I2C bus speed | 100 kHz to 400 kHz | Supported operating range |
| I2C addressing | 7-bit slave | Valid configurable addresses are `0x08..0x77` |
| Factory I2C address | `0x20` | Pass the 7-bit value, not wire byte `0x40` |
| Device protocol | DDP v1.0 | Command transaction followed by an exact-length read transaction |
| Logical Device ID | `0x0102` | TEMT6000 identity; independent of I2C address |
| Firmware / hardware | 1.0 / 1.0 | Current observable controller profile |
| Capability bitmap | `0x000001B9` | I2C config, analog input, sensor data, relay, watchdog, persistent config |
| Raw digital sample | `0` to `4095` | 12-bit ADC code returned as unsigned 16-bit little endian |
| ADC update interval | Approximately 20 ms | Background acquisition; I2C read returns latest published sample |
| Command processing delay | 2 to 5 ms | Use 5 ms conservatively before reading |
| Pending setter timeout | 250 ms | Parameter must arrive before expiry |
| Direct `SIG` range | Not specified | Requires current analog-stage schematic and measurement |
| Module current consumption | Not specified | Controller, LEDs, pull-ups, and sensor load are undocumented |
| Module ambient temperature | Not specified | No complete-module qualification supplied |

The controller's 2.0–5.5 V rating confirms that the controller itself can operate at 5 V; it does not by itself validate 5 V operation for every circuit on the module. The direct `SIG` path and I2C levels are referenced to the powered system, so every attached host must tolerate the resulting voltage or use level translation. Complete-module absolute maximums, pull-up resistance, and current consumption still require the schematic and qualification.

## 2.2 Interface Controller Characteristics

The supplied UE0102 reference manual identifies a PY32F003L24D6TR and publishes 1.7–5.5 V and −40 to +85 °C for that separate development board. It does not establish which ordering suffix was procured for this UE0098 assembly. This Product Reference therefore does not identify the fitted ordering suffix; it uses the x7 range of 2.0–5.5 V as conservative voltage guidance.

| Feature | Controller capability |
|---|---|
| CPU and application clock | 32-bit Arm Cortex-M0+; internal HSI at up to 24 MHz; external HSE oscillator not used |
| Memory | 16 KB Flash and 2 KB SRAM |
| Published operating voltage | 2.0–5.5 V, conservative x7 range |
| ADC | One 12-bit ADC, up to 10 external channels, conversion range `0..VCC` |
| I2C | Standard mode 100 kHz, Fast mode 400 kHz, 7-bit addressing |
| GPIO | Up to 18 I/Os, all available as external interrupts |
| Timers | TIM1, TIM3, TIM14, TIM16, TIM17, LPTIM, IWDG, WWDG, SysTick, IRTIM |
| Other interfaces | SPI, two USARTs, DMA, RTC, CRC-32, two comparators, UID; SWD is reserved for factory use on this module |

The controller range applies to the controller only. It does not establish the pull-up resistance, complete-module current, or absolute maximum of every external contact.

## 2.3 TEMT6000 Maximum Ratings

The values below are included only as a comparative TEMT6000 profile. They are not proof that a Vishay part is fitted, are not guaranteed for the fitted sensor, and do not define limits for the controller, LEDs, pull-ups, connectors, or complete board.

| Parameter | Symbol | Value | Unit |
|---|---:|---:|---|
| Collector-emitter voltage | `VCEO` | 6 | V |
| Emitter-collector voltage | `VECO` | 1.5 | V |
| Collector current | `IC` | 20 | mA |
| Power dissipation at 25 °C | `PV` | 100 | mW |
| Junction temperature | `Tj` | 100 | °C |
| Component operating temperature | `Tamb` | −40 to +100 | °C |

## 2.4 TEMT6000 Characteristics

Unless noted otherwise, the values are specified at 25 °C and are provided for comparison only. Production specifications must come from the confirmed fitted part and module-level validation.

| Parameter | Test condition | Min. | Typ. | Max. | Unit |
|---|---|---|---|---|---|
| Collector dark current | `VCE = 5 V`, `Ev = 0` | — | 3 | 50 | nA |
| Collector light current | `Ev = 20 lx`, CIE illuminant A, `VCE = 5 V` | 3.5 | 10 | 16 | µA |
| Collector light current | `Ev = 100 lx`, CIE illuminant A, `VCE = 5 V` | — | 50 | — | µA |
| Collector-emitter capacitance | `VCE = 0 V`, `f = 1 MHz`, dark | — | 16 | — | pF |
| Collector-emitter saturation voltage | `Ev = 20 lx`, `IPCE = 1.2 µA` | — | 0.1 | — | V |
| Angle of half sensitivity | — | — | ±60 | — | degrees |
| Peak sensitivity wavelength | — | — | 570 | — | nm |
| Spectral bandwidth at half sensitivity | — | 440 | — | 800 | nm |

## 2.5 Legacy Analog Circuit Scope

The V0.0.1 schematic shows a TEMT6000 with a 10 kΩ emitter resistor, giving the first-order relation `VSIGNAL ≈ IPCE × 10 kΩ` outside saturation. Applying the 100 lx typical current from the datasheet predicts about 0.50 V; this is not a guaranteed current-board value.

V0.3.1 adds an interface controller and other circuitry. Without its schematic, the legacy resistor value and transfer function must not be represented as a guaranteed current-board characteristic. Direct `SIG` must be measured and validated on V0.3.1.

## 2.6 Unspecified Current-Module Characteristics

- Complete-module absolute maximum, current consumption, and power-up behavior
- Exact oscillator/timing tolerance and electrical bus-loading limits
- Pull-up resistance, bus capacitance allowance, and level compatibility
- Direct analog transfer function, load resistance, range, accuracy, and source impedance
- Guaranteed lux range, accuracy, repeatability, response time, and calibration
- Exact controller package/ordering suffix, released firmware image, and update procedure
- Board-level temperature, ESD, EMC, humidity, and ingress ratings

## 2.7 Electrical Precautions

1. Use a current-limited 3.3 V or 5 V supply for engineering bring-up.
2. Verify connector orientation and share ground before applying power.
3. Do not exceed the controller upper operating limit of 5.5 V.
4. At 5 V, confirm host tolerance at `SDA`, `SCL`, and `SIG`, or add suitable level translation before connection.
5. Do not connect to reset or SWD; these signals are reserved for manufacturer programming and advanced factory diagnostics, not user firmware replacement.
6. Remove power before cutting or reworking the I2C solder bridge.
7. Confirm the current schematic and interface specification before production.

# 3 Functional Overview

The TEMT6000 sensor generates a light-dependent analog signal. That signal is exposed at `SIG` and acquired by the onboard controller through `PA2/ADC0` for DDP access. The controller uses its internal 24 MHz clock; this application does not use an external oscillator. Controller output PB5 drives the `BUILTIN` LED.

The missing schematic must still confirm the passive sensor circuitry and complete module electrical characteristics. The logical associations and observable firmware behavior documented below are the released interface.

## 3.1 Functional Block Diagram

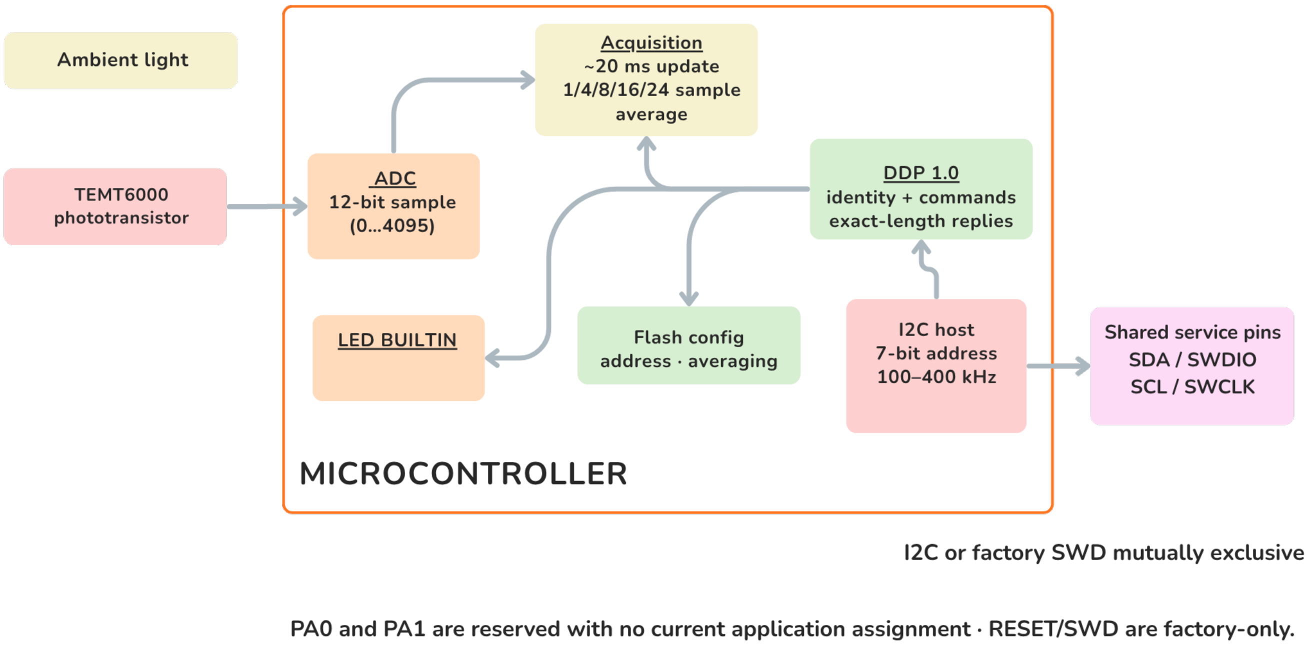



**Figure 3.1 — Functional signal and control paths.** The direct `SIG` branch and controller PA2 ADC observe the sensor path. Firmware averages the sample and publishes it through DDP over I2C. The internal HSI supplies the 24 MHz application clock; no HSE crystal or external clock is used.

## 3.2 Controller Clock and Execution Model

| Property | Application configuration | Functional consequence |
|---|---|---|
| Controller core | 32-bit Arm Cortex-M0+ | Executes acquisition, DDP, configuration, and indicator control |
| Controller memory | 16 KB Flash and 2 KB SRAM | Memory configuration for this module |
| Maximum application clock | 24 MHz | Maximum clock used and documented for this product firmware |
| System-clock source | Internal high-speed oscillator (HSI) | Self-clocked controller operation |
| External high-speed clock (HSE) | Not used | No external oscillator or crystal is required by this application |
| ADC execution | Background task, approximately every 20 ms | Reads return the most recently published sample rather than starting a conversion |
| Command execution | Command is prepared after an I2C write | Host waits 2–5 ms before the exact-length read |

The 24 MHz value describes this application configuration. It must not be replaced by a higher generic family maximum when specifying this product.

## 3.3 Measurement and Command Flow

1. Ambient light changes the TEMT6000 analog output.
2. Controller input `PA2/ADC0` performs a 12-bit conversion.
3. Firmware discards one settling conversion, updates the circular sample buffer, and publishes the rounded average.
4. A host addresses the module and writes one DDP command byte.
5. After STOP and a 2–5 ms processing interval, the host reads the exact response length.
6. `READ_ADC0` and `TEMT6000_RAW` return the same latest published sample.

| Functional timing | Value | Notes |
|---|---:|---|
| ADC update interval | Approximately 20 ms | About 50 background updates per second |
| Averaging selections | 1, 4, 8, 16, or 24 samples | Persistent configuration |
| Full averaging-window fill | `N × 20 ms` | 20, 80, 160, 320, or 480 ms |
| I2C bus clock | 100 kHz to 400 kHz | Standard mode and Fast mode |
| Command-to-read delay | 2–5 ms | Use 5 ms conservatively |
| Staged-parameter timeout | 250 ms | Applies to `0x21`, `0x62`, and `0xA3` |

## 3.4 Logical Register Association Map

DDP presents command-selected data fields rather than MCU memory addresses. The following table associates each externally visible logical field with its physical resource and command address.

| Logical field | Physical association | DDP command address | Data or behavior |
|---|---|---:|---|
| Device identity | Firmware constant | `0x00` | Device ID `0x0102` |
| Firmware version | Firmware metadata | `0x01` | Version 1.0 |
| Hardware version | Product profile | `0x02` | Version 1.0 |
| Capabilities | Firmware feature bitmap | `0x03` | `0x000001B9` |
| Protocol version | DDP implementation | `0x04` | DDP 1.0 |
| Active I2C address | I2C slave configuration | `0x20` | Factory `0x20`; configurable `0x08..0x77` |
| Persistent configuration | Internal Flash | `0x21..0x24`, `0x62`, `0xA3` | Address, averaging, and toggle time |
| `ADC0` | TEMT6000 signal on controller PA2 | `0x60` | Unsigned 12-bit ADC code |
| ADC averaging | Circular sample buffer | `0x62`, `0x63` | 1, 4, 8, 16, or 24 samples |
| TEMT6000 sensor data | Same published `PA2/ADC0` sample | `0x80` | Same two-byte response as `0x60` |
| Built-in indicator | Controller `PB5/BUILTIN` | `0xA0..0xA4` | Off, on, toggle, and persistent pulse time |
| System reset | Controller reset path | `0xF0` | Resets without a response read |
| I2C service pins | `PA10/SDA`, `PB6/SCL` | Transport, not a data register | Shared with factory-only SWD aliases |
| Reserved contacts | `PA0`, `PA1` | None | No current application assignment |

## 3.5 I2C and DDP Register Model

DDP is **not** a memory-mapped register interface. The first byte written after the 7-bit slave address is a command selector analogous to a register address. The command byte selects a prepared response or an operation; it does not expose the PY32F003 memory map.

The bus address and command address are independent values. A host passes the **7-bit I2C slave address** (`0x20` at the factory) to its library, then writes one **DDP command address** from `0x00` through `0xFF`. Do not pass shifted wire bytes `0x40`/`0x41` as the device address.

```
host → [7-bit slave address + W] [DDP command address] → STOP
       wait 2…5 ms
host ← [7-bit slave address + R] [exact response length] ← STOP
```

Multi-byte values are little endian. Staged setters require a second one-byte write transaction within 250 ms; the selector and parameter must not be sent in the same I2C write transaction.

## 3.6 Global Command Address Map

| DDP address block | Commands defined for this profile | Current support |
|---|---|---|
| 0x00..0x1F Device information | 0x00..0x04 | Identity and discovery registers implemented |
| 0x20..0x3F Configuration | 0x20..0x24 | I2C address and persistent configuration implemented |
| 0x40..0x5F Digital I/O | None | Not implemented on this module |
| 0x60..0x7F Analog input | 0x60, 0x62, 0x63 | ADC0/PA2 and averaging implemented; 0x61 unsupported |
| 0x80..0x9F Sensor data | 0x80 | TEMT6000 raw sample implemented |
| 0xA0..0xBF Actuators | 0xA0..0xA4 | PB5/BUILTIN indicator control implemented |
| 0xC0..0xDF Calibration | None | Capability not announced; unsupported |
| 0xE0..0xEF Reserved | None | Reserved; do not use |
| 0xF0..0xFF System | 0xF0..0xF4 | Reset implemented; watchdog experimental; remaining functions incomplete or unsupported |

Any undefined or unsupported selector returns the current packed unknown-command response with low nibble 0xB.

## 3.7 Command Register Descriptions

Access notation: R selects and reads a prepared response; W performs an operation or begins a staged parameter write; W/R writes the selector and then reads its acknowledgement or result.

### 3.7.1 Device Information Registers

| Address | Symbol | Access | Exact response | Association |
|---|---|---|---|---|
| 0x00 | GET_DEVICE_ID | R | [0x02, 0x01] | Device ID 0x0102 |
| 0x01 | GET_FIRMWARE_VERSION | R | [0x01, 0x00] | Firmware major 1, minor 0 |
| 0x02 | GET_HARDWARE_VERSION | R | [0x01, 0x00] | Hardware major 1, minor 0 |
| 0x03 | GET_CAPABILITIES | R | [0xB9, 0x01, 0x00, 0x00] | Little-endian bitmap 0x000001B9 |
| 0x04 | GET_PROTOCOL | R | [0x01, 0x00] | DDP major 1, minor 0 |

Discovery order is 0x04, 0x00, 0x01, 0x02, then 0x03. A host must verify protocol major 1 and Device ID 0x0102; the I2C address alone does not identify the product.

### 3.7.2 Configuration Registers

| Address | Symbol | Access | Parameter or response | Association |
|---|---|---|---|---|
| 0x20 | GET_I2C_ADDR | R | One-byte active 7-bit address | I2C peripheral address |
| 0x21 | SET_I2C_ADDR | W | Staged 0x08..0x77; ACK low nibble 0xD | Saves Flash and resets after final ACK |
| 0x22 | SAVE_CONFIG | W/R | Returns 0x00 | Idempotent; setters already persist |
| 0x23 | RESET_FACTORY | W/R | ACK low nibble 0xE | Restores defaults; separate reset required |
| 0x24 | GET_I2C_STATUS | R | 0x00 default; 0x01 loaded from Flash | Address-source status |

### 3.7.3 Input and Sensor Registers

| Address | Symbol | Access | Parameter or response | Physical association |
|---|---|---|---|---|
| 0x60 | READ_ADC0 | R | [LSB, MSB], valid code 0..4095 | Latest PA2 sample |
| 0x62 | SET_ADC_AVERAGING | W | Staged 1, 4, 8, 16, or 24; ACK low 0xC | Persistent averaging depth |
| 0x63 | GET_ADC_AVERAGING | R | One byte: 1, 4, 8, 16, or 24 | Active averaging depth |
| 0x80 | TEMT6000_RAW | R | [LSB, MSB], valid code 0..4095 | Same published sample as 0x60 |

READ_GPIO0/1 (0x40/0x41), WRITE_GPIO0/1 (0x42/0x43), and READ_ADC1 (0x61) are unsupported.

### 3.7.4 Indicator and System Registers

| Address | Symbol | Access | Parameter, response, or effect | Physical association |
|---|---|---|---|---|
| 0xA0 | RELAY_OFF | W/R | ACK low nibble 0x0; output LOW | PB5/BUILTIN off |
| 0xA1 | RELAY_ON | W/R | ACK low nibble 0x1; output HIGH | PB5/BUILTIN on |
| 0xA2 | RELAY_TOGGLE | W/R | ACK low nibble 0x6; non-blocking pulse | PB5/BUILTIN toggle |
| 0xA3 | SET_TOGGLE_TIME | W | Staged 1..40; ACK low nibble 0x7 | Persistent units of 25 ms |
| 0xA4 | GET_TOGGLE_TIME | R | One byte 1..40 | Active pulse time |
| 0xF0 | RESET | W | Immediate reset; do not request a reply | Controller reset path |

| Address | Symbol | Access | Parameter, response, or effect | Physical association |
|---|---|---|---|---|
| 0xF1 | WATCHDOG_RESET | W | Experimental; no active hardware IWDG | Firmware system block |
| 0xF2..0xF4 | Reset-info/NRST functions | — | Unsupported or incomplete | Reserved for future profile completion |

## 3.8 Response-Byte Interpretation

Response bytes have meaning only within the selected command and expected length. For example, `0x01` may mean firmware major 1, hardware major 1, protocol major 1, the high byte of Device ID `0x0102`, or “address loaded from Flash” from `GET_I2C_STATUS`. It is not a universal success code.

## 3.9 Direct Analog Mode

The top-side `SIG` contact provides direct sensor access for ADC experiments, production test, or calibration. Its transfer function and safe input range are not documented. Configure the host as a high-impedance analog input and measure the actual range before connecting a lower-voltage ADC.

The controller samples this sensor path internally on `PA2`, mapped to logical `ADC0`. `READ_ADC0` and `TEMT6000_RAW` are equivalent in the current firmware.

## 3.10 I2C Disable Bridge

The released pinout instructs the user to cut a solder bridge to disable I2C. The missing current schematic leaves the exact isolation boundary unspecified. Do not assume that cutting the bridge removes pull-ups, disconnects every controller pin, or changes the analog transfer in a particular way. Rework only with power removed and after continuity checks on the target revision.

## 3.11 Optical Response

Reference-only TEMT6000X01 data lists a 570 nm peak, a 440 nm to 800 nm half-sensitivity spectral bandwidth, and a ±60° half-sensitivity angle. These values are comparative, not guaranteed for the unconfirmed fitted part. Source spectrum, temperature, sensor spread, enclosure windows, and mechanical shadowing can also change the response.

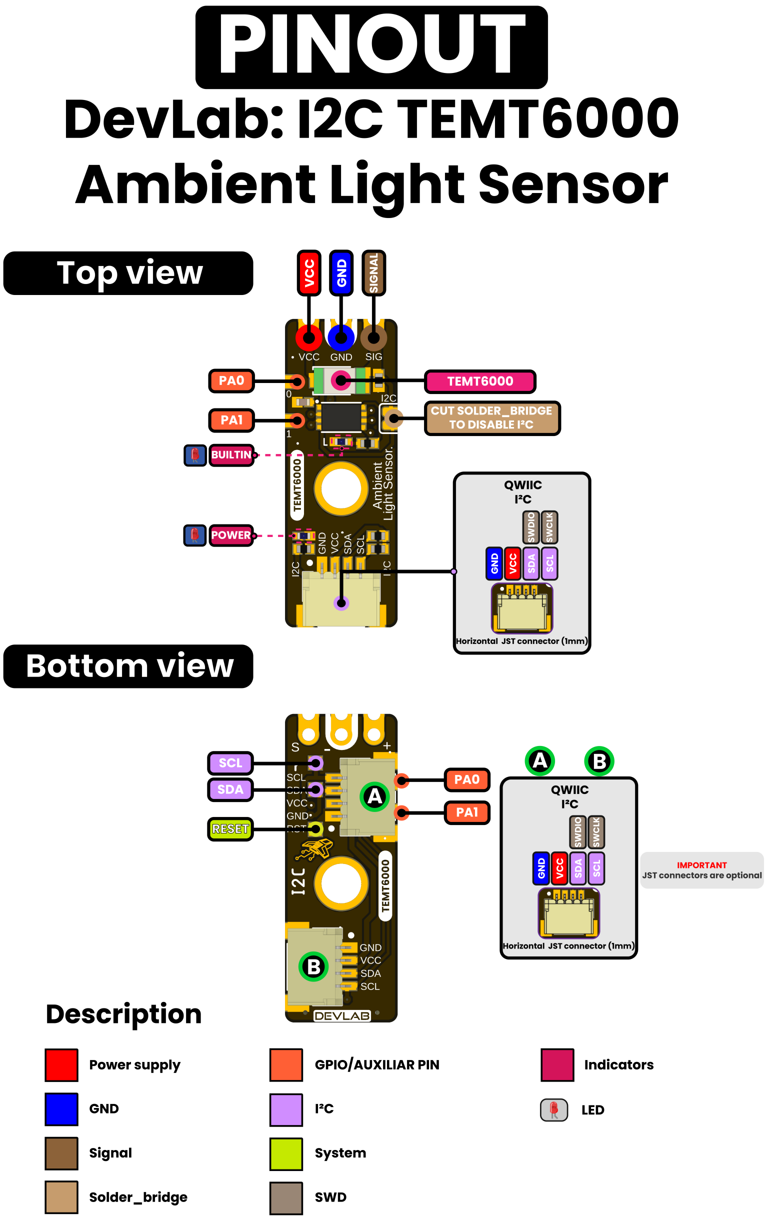
## 3.12 Service and Expansion Signals

The pinout exposes `PA0`, `PA1`, `RESET`, `SWDIO`, and `SWCLK`. `PA0` and `PA1` have no application assignment and are reserved for future use. `SWDIO` and `SWCLK` are alternate functions of the same `PA10`/`PB6` pins used for `SDA`/`SCL`, not a separate port. They and `RESET` are reserved for manufacturer programming and advanced factory diagnostics; user firmware replacement is unsupported.

# 4 Connectors & Pinouts

Pin labels below come from the released English pinout. It does not provide controlled contact numbers or a mating-connector part number, so the signal order must be checked against connector orientation before producing a harness.

## 4.1 General Pinout




## 4.2 Direct Sensor Contacts

| Label | Type | Description |
|---|---|---|
| VCC | Power | Module supply; nominal 3.3 V or 5 V operation |
| GND | Power | Common power and signal reference |

| Label | Type | Description |
|---|---|---|
| `SIG` / `SIGNAL` | Analog | Direct light-dependent sensor signal; current transfer function unspecified |

These three contacts are shown at the end opposite the primary Qwiic connector.

## 4.3 Qwiic I2C Positions A and B

| Signal | Type | Description |
|---|---|---|
| `GND` | Power | Common return |
| `VCC` | Power | I2C peripheral supply; nominal 3.3 V or 5 V |
| `SDA` / `SWDIO` | Bidirectional I2C / factory debug data | One physical `PA10` line; normal use is SDA; host must tolerate the bus pull-up voltage |
| `SCL` / `SWCLK` | I2C clock / factory debug clock | One physical PB6 line; normal use is SCL at 100 kHz to 400 kHz |

The pinout shows two possible horizontal 1.0 mm JST connector positions and marks JST connectors as optional. Verify which position is populated on the ordered assembly. Parallel connectors are intended for bus pass-through, but that topology must be confirmed by the current schematic.

I2C and SWD are multiplexed on this same physical port: `PA10` is `SDA`/`SWDIO`, and PB6 is `SCL`/`SWCLK`. They are mutually exclusive, and there is no independent SWD connector or pin pair.

## 4.4 Auxiliary and Factory Service Functions

| Label | Documented role | Qualification status |
|---|---|---|
| `PA0` | GPIO / reserved pin | No current application assignment; electrical limits unspecified |
| `PA1` | GPIO / reserved pin | No current application assignment; electrical limits unspecified |
| `RESET` | System reset | Factory service only; not intended for routine user operation |

Do not connect an SWD probe or drive reset/debug functions. They are not user interfaces, and the product is not designed for user firmware replacement. Manufacturer diagnostics must make I2C inactive and isolate other bus devices before selecting SWD on the shared pins.

## 4.5 Indicators and I2C Bridge

The pinout identifies a `POWER` LED, a firmware-controlled `BUILTIN` LED, and a cuttable bridge used to disable I2C. Controller pin PB5 drives `BUILTIN`. Indicator polarity/current and bridge net connections still require the current schematic. The sensor ADC input is mapped internally to controller PA2; it is not an additional external contact in this pinout.

# 5 Board Operation

The repository supports safe electrical bring-up through I2C scanning, DDP identity and measurement reads, and direct analog observation.

## 5.1 I2C Bring-up

1. Inspect the board revision, connector population, and I2C bridge.
2. With power off, select a nominal 3.3 V or 5 V supply and connect GND, VCC, SDA, and SCL in the documented orientation. For 5 V operation, verify host tolerance at the bus pull-up voltage or add level translation.
3. Power from a current-limited source and check for unexpected heating.
4. Select an I2C clock from 100 kHz through 400 kHz; maintained examples use 400 kHz. The ESP32 and MicroPython examples map host GPIO6 to SDA and GPIO7 to SCL. Start at factory 7-bit address `0x20`, or scan if it was changed.
5. Record the active address, protocol, firmware, hardware, and capability values reported during discovery.
6. Verify protocol/firmware/hardware 1.0 and capabilities `0x000001B9`, then read either `READ_ADC0` (`0x60`) or equivalent `TEMT6000_RAW` (`0x80`).
7. Use a logic analyzer to capture start, address, acknowledge, STOP, and read timing if enumeration fails.

The maintained DDP clients are `software/examples/i2c/cpp_examples/ddp_temt6000/ddp_temt6000.ino` and `software/examples/i2c/micropython/ddp_temt6000.py`. The MicroPython example imports the reusable `software/examples/i2c/micropython/lib/devlab_ddp.py` module, which is installed as `/lib/devlab_ddp.py` on the host. Low-level scanner sketches are `software/examples/i2c/cpp_examples/i2c_scanner/i2c_scanner.ino` and `software/examples/i2c/micropython/i2c_scan.py`.

## 5.2 Application 1: Read Ambient-Light Data

1. Initialize I2C at 100 kHz or 400 kHz using 7-bit address `0x20`.
2. Run the Chapter 3 discovery sequence and accept only DDP major 1 with Device ID `0x0102`.
3. Write `TEMT6000_RAW` (`0x80`), issue STOP, wait 5 ms, and read exactly two bytes.
4. Decode `raw = response[0] | (response[1] << 8)` and reject values above 4095.
5. Repeat no faster than the approximately 20 ms background update interval when a new sample is required.

`READ_ADC0` (`0x60`) may replace `TEMT6000_RAW`; both access the same published `PA2/ADC0` sample. The value is an uncalibrated ADC code, not lux. Convert to lux only after calibration against a reference meter in the final optical and mechanical installation.

## 5.3 Application 2: Change the I2C Address

The common DDP block provides `GET_I2C_ADDR` (`0x20`), `SET_I2C_ADDR` (`0x21`), `SAVE_CONFIG` (`0x22`), `RESET_FACTORY` (`0x23`), and `GET_I2C_STATUS` (`0x24`). Factory address is `0x20`; a new value must be within `0x08..0x77`.

Address change is a staged operation. Send `0x21`, wait/read ACK low nibble `0xD`, send the new one-byte value within 250 ms, wait about 100 ms, and read the final ACK at the old address. The setter saves Flash and reset follows when the final ACK is consumed. Wait about 300 ms and identify `0x0102` at the new address. `SAVE_CONFIG` is currently idempotent. Factory restore returns ACK low `0xE` but requires a separate reset before address `0x20` becomes active.

## 5.4 Application 3: Averaged Capture

The controller updates ADC0 in the background about every 20 ms and publishes a rounded circular-buffer mean. `GET_ADC_AVERAGING` (`0x63`) returns 1, 4, 8, 16, or 24. `SET_ADC_AVERAGING` (`0x62`) is staged with ACK low nibble `0xC` and persists to Flash. Read before writing, and wait `N × 20 ms` for a full window. Recommended values are 8 for UI/general automation and 16 for stable ambient measurement. Do not poll faster than 20 ms expecting a new value.

## 5.5 Application 4: Built-in Indicator

Controller `PB5/BUILTIN` uses the relay-compatible commands: `RELAY_OFF` (`0xA0`), `RELAY_ON` (`0xA1`), and non-blocking `RELAY_TOGGLE` (`0xA2`). It is not a digital GPIO signal, and the digital-I/O commands are unsupported. Pulse time is configured with `SET_TOGGLE_TIME` (`0xA3`) in `1..40` units of 25 ms.

## 5.6 Application 5: Direct Analog Bring-up

1. Leave the I2C/service pins undriven and configure a host ADC as a high-impedance input.
2. Connect the direct `GND`, `VCC`, and `SIG` contacts with power removed.
3. Verify the actual `SIG` range using a multimeter before relying on the ADC; the controller ADC conversion range is `0..VCC`.
4. Compare covered and illuminated readings and check that the response is repeatable.
5. Calibrate against a reference meter for quantitative lux measurements.

The analog examples remain available under `software/examples/adc/`, separate from the I2C clients. Their voltage calculations require measurement because the only available schematic describes legacy V0.0.1 hardware.

## 5.7 Service Operation: Modifying the I2C Bridge

Do not cut the bridge as a first troubleshooting step. First verify power, connector order, pull-ups, bus ownership, and scan timing. If analog-only operation requires isolation, remove power, document the original bridge, cut only the marked feature, inspect for debris, and confirm continuity before repowering. Restoring the bridge requires controlled solder rework.

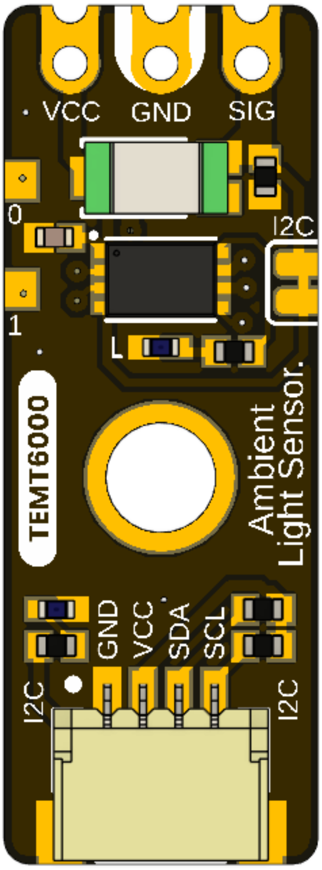
## 5.8 Troubleshooting Guide

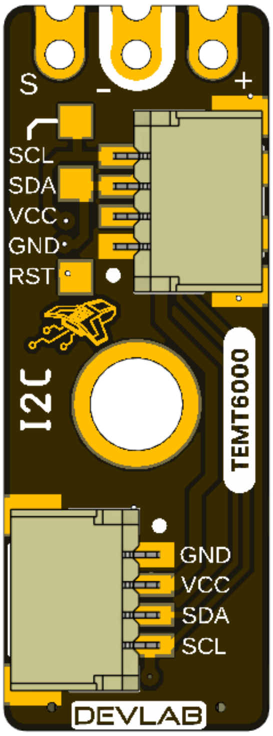
| Symptom | Checks |
|---|---|
| No address found | Supply/bus-logic compatibility, ground, connector orientation, populated connector, SDA/SCL continuity, intact I2C bridge |
| Bus held low | Cable reversal, duplicate pull-ups, unintended drive on shared `SDA/SWDIO` or `SCL/SWCLK`, solder bridge, unpowered device |
| Address found but rejected | Verify DDP major 1, Device ID `0x0102`, exact response lengths, and possible address collision |
| DDP identity succeeds but measurement fails | Check `ANALOG_INPUT/SENSOR_DATA`, 5 ms delay, exact length, byte order, and 12-bit range |
| Direct `SIG` always zero/full scale | Supply, ADC range, contact order, optical obstruction, current-board transfer function |
| Built-in LED unexpected | Use relay-compatible commands for PB5; digital-I/O commands are unsupported |
| Lux result inaccurate | Sensor spread, source spectrum, geometry, enclosure, temperature, ADC and calibration |

# 6 Mechanical Information

The current package provides rendered top and bottom views for V0.3.1 but no controlled current-revision dimension drawing. Legacy V0.0.1 dimensions are archived under `hardware/resources/old/` and describe a materially different, shorter analog-only board.

## 6.1 Current Board Envelope






The board has a narrow module outline, direct contacts at one end, at least one central mounting feature, and optional horizontal Qwiic connector positions. Rendered views are suitable for identification, not production measurement.

## 6.2 Integration Clearances

Leave optical clearance around the transparent TEMT6000 sensor package and its field of view. Provide connector insertion and cable bend clearance at every populated Qwiic position. Keep conductive fasteners away from pads and service signals, and avoid enclosure features that shadow the sensor or LEDs.

## 6.3 Mechanical Data Not Specified

- Board length, width, thickness, tolerances, and finished mass
- Hole diameters, coordinates, and edge clearances
- Optional connector population variants and mating-part numbers
- Connector insertion envelope, retention force, and cable bend radius
- Maximum top- and bottom-side component heights
- Optical axis, keep-out cone, and enclosure-window specification
- PA0/PA1 and service-pad coordinates

Do not scale production dimensions from the images or reuse the archived V0.0.1 mechanical values for V0.3.1.

# 7 Company Information

| Item | Information |
|---|---|
| Company / brand | UNIT Electronics |
| Board ecosystem | DevLab |
| Product family | Atom |
| Website | UNIT Electronics |
| Address | Salvador 19, Cuauhtémoc, 06000 Mexico City, CDMX, Mexico |

The product repository and controlled technical references are listed in the next chapter.

# 8 Reference Documentation

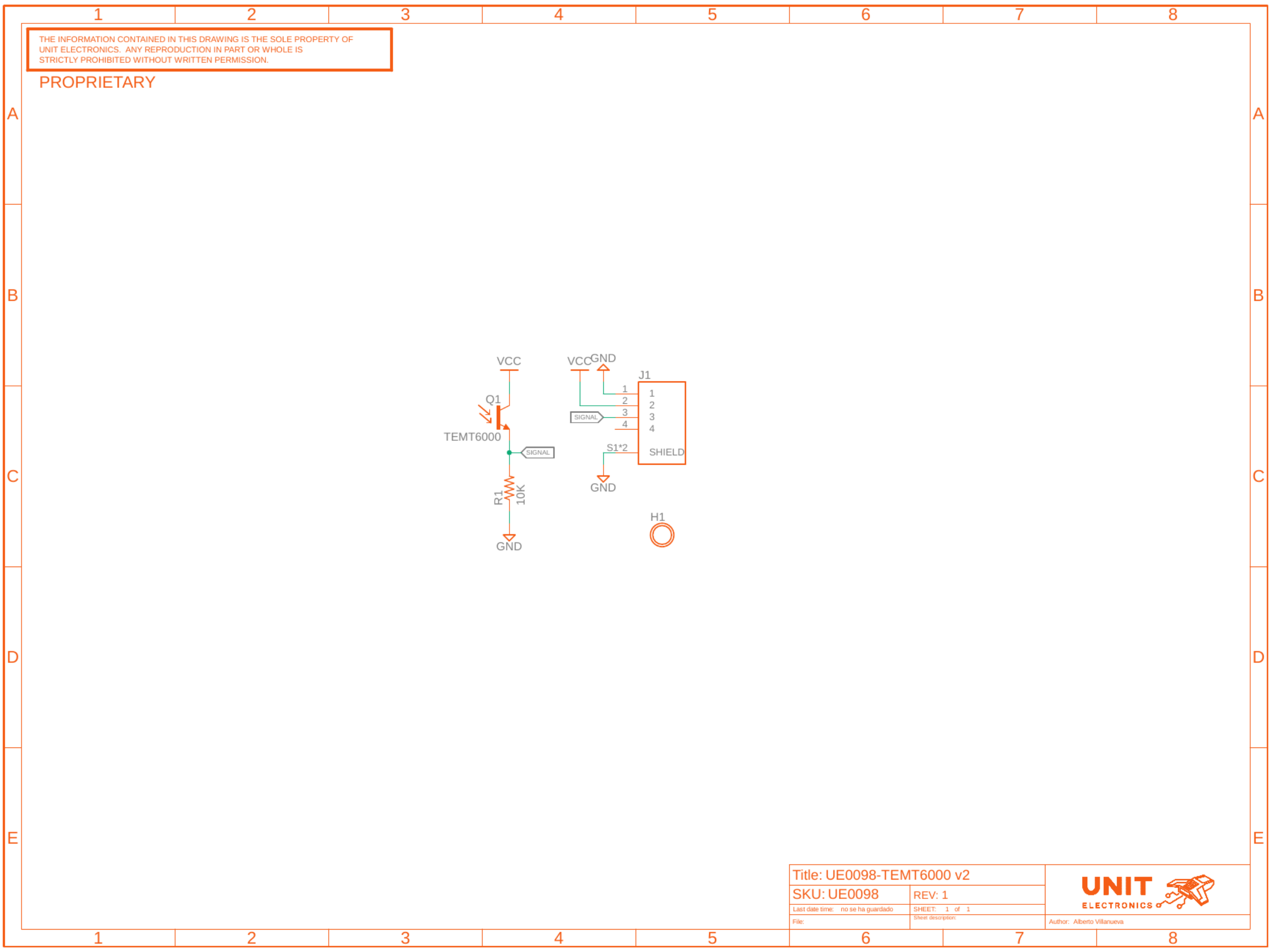
| Reference | Location |
|---|---|
| Product repository | UNIT-Electronics-MX/unit_devlab_temt600_ambient_light_sensor |
| Hardware reference | hardware/README.md |
| English pinout | Released pinout — English |
| Spanish pinout | Released pinout — Spanish |
| Board top view | V0.3.1 resource |
| Board bottom view | V0.3.1 resource |
| Legacy schematic | Analog V0.0.1 |
| Software examples | C++ and MicroPython |
| Local DDP device profile | TEMT6000 DDP profile |
| Canonical DDP definitions and Arduino library | UNIT-Electronics-Labs/unit_devlab_ddp_library |
| Comparative component page | Vishay TEMT6000X01 — reference only |
| Comparative component datasheet | Vishay Semiconductors, *TEMT6000X01 Ambient Light Sensor*, document 81579 — reference only |

The released pinout and current board images define the visible interfaces. The legacy schematic defines only the earlier analog module. The Vishay datasheet is retained only as a comparative TEMT6000X01 profile; it does not establish the manufacturer or orderable part fitted to this module and does not define complete-module ratings. The DDP profile defines runtime identity and digital data transport; it does not replace the missing V0.3.1 electrical specification.

# 9 Appendix

## 9.1 Legacy V0.0.1 Schematic

The image below is rendered from `hardware/unit_sch_V_0_0_1_ue0098_TEMT6000.pdf`. It documents the former analog-only module and is retained for traceability. It is **not** the electrical schematic for the controller-based V0.3.1 board.




The legacy circuit connects a TEMT6000 phototransistor and 10 kΩ load to one analog signal. It contains no interface controller, SDA, SCL, PA0, PA1, reset, SWD, indicators, or I2C disable bridge.

## 9.2 Document Control

| Field | Value |
|---|---|
| Product | UNIT ATOM TEMT6000 Ambient Light Sensor |
| Product hierarchy | UNIT Electronics → DevLab board ecosystem → Atom family |
| Manufacturer Part Number (MPN) | UE0098 |
| Current board artwork | V0.3.1 |

| Field | Value |
|---|---|
| Current pinout | V3.1.0 |
| Available schematic | Legacy V0.0.1 only |
| Product Reference | Version 1.1.0 |
| Publication date | 2026-08-31 |
| Interfaces | Qwiic I2C with shared factory SWD, analog, and auxiliary pads |

## 9.3 Required Technical Releases

- V0.3.1 schematic and bill of materials
- Exact controller package/ordering suffix, released firmware image, and update procedure
- Exact oscillator/timing tolerances and uniform future DDP status behavior
- Complete-module supply/logic absolute-maximum ratings and current consumption
- Pull-up values and bus-loading limits
- Direct `SIG` transfer, loading, range, and ADC guidance
- Current-revision controlled mechanical drawing
- Module-level optical, electrical, environmental, and EMC characterization

## 9.4 Source Notes and Inconsistencies

- Current board images and pinout resources use different revision-number formats in their filenames and storage directories.
- Current pinout artwork calls the product `DevLab: I2C TEMT6000`; this reference identifies UNIT Electronics as the company that creates and develops the DevLab board ecosystem, with Atom as a product family within it.
- The current pinout shows an onboard controller and I2C/debug resources, but the only schematic in the repository is the earlier analog V0.0.1 circuit.
- Archived V0.0.1 dimensions, topology, and board views do not describe the longer V0.3.1 controller-based board.
- Qwiic positions A and B are marked optional; released assembly variants and controlled contact numbering are not supplied.
- Earlier examples treated the sensor as a digital threshold input. The sensor signal is analog; I2C operation is provided by the new onboard controller.
- Current firmware maps PA2 to `ADC0` and PB5 to the relay-compatible actuator block. Digital-I/O commands `0x40..0x43` are not implemented; exposed `PA0`/`PA1` remain unassigned.

## 9.5 Current Firmware Limitations

- The `WATCHDOG` capability is announced, but the current watchdog command does not operate an active hardware IWDG and must remain experimental.
- `GET_RESET_INFO` is not implemented and NRST behavior is incomplete.
- An internal ADC error can be returned as `0x0FFF`, indistinguishable from a legitimate full-scale sample.
- Current main-loop code does not call the available prolonged-BUSY recovery.
- There is no sample timestamp, sample counter, or calibrated lux result.